

2020

Full Marks : 70

Time : 3 hours

The figures in the right-hand margin indicate marks

Answer from **all** the Parts as directed

Part—A

(Objective Type Questions)

1. Choose the correct answer in each of the following : 5×1

(a) The length of the arc of the curve $y = \log \sec x$ between $x = 0$ and $x = \frac{\pi}{6}$ is equal to

- (i) $\log 3$ (ii) $2 \log 3$
(iii) $\frac{1}{2} \log 3$ (iv) None of these

(b) $\int \frac{dx}{x^2 + x + 1}$ is equal to

(i) $\frac{1}{2} \log \left| \frac{x-1}{x+1} \right|$

(ii) $\frac{1}{\sqrt{2}} \log \left| \frac{x-\sqrt{2}}{x+\sqrt{2}} \right|$

(iii) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{3} \right)$

(iv) None of these

(c) What is DES ?

(i) ~~Block cipher~~

(ii) Stream cipher

(iii) Byte cipher

(iv) None of these

(d) Moment of inertia of a circular plate about a line perpendicular to the plate through the centre is

(i) Ma^2 (ii) ~~$\frac{1}{2} Ma^2$~~

(iii) $\frac{1}{3} Ma^2$ (iv) $\frac{1}{4} Ma^2$

(e) The degree of differential equation satisfying

$$\left(\frac{d^2 y}{dx^2} \right)^{3/2} - \left(\frac{dy}{dx} \right)^{1/2} - 4 = 0$$

is

(i) 6 (ii) 3

(iii) 2 (iv) ~~4~~

2. Fill in the blanks : 5×1

(a) In cryptography the order of the letter in a message is rearranged by

Transposition.

(b) The equation $y = px + f(p)$ is known as _____.

(c) Integration

$$\int \frac{dx}{\sqrt{(x-2)(x-3)}}$$

is _____.

(d) Integrating factor of the differential equation

$$\cos^2 x \frac{dy}{dx} + y = \tan x$$

is $\tan x$.

(e) The linear function of the variables which is to be maximize or minimize is called _____.

Part—B

(Short Answer Type Questions)

Answer any four questions of the following : 4×5

3. (a) Define hyperplane. Show that a hyperplane is a convex set.

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(Turn Over)

- (b) Solve the following linear programming problem by graphical method :

$$\text{Max } Z = 5x_1 + 7x_2$$

subject to

$$x_1 + x_2 \leq 4$$

$$3x_1 + 8x_2 \leq 24$$

$$10x_1 + 7x_2 \leq 35$$

$$x_1, x_2 \geq 0$$

4. (a) Find the limit of

$$\frac{n}{n^2 + 1^2} + \frac{n}{n^2 + 2^2} + \dots + \frac{n}{n^2 + n^2}$$

as $n \rightarrow \infty$.

- (b) Evaluate :

$$\int_0^{\pi/2} \sin^4 x \cos^4 x \, dx$$

5. (a) Solve :

$$(1 + y^2) \, dx = (\tan^{-1} y - x) \, dy$$

- (b) Find the orthogonal trajectories of $r\theta = a$.

6. (a) Give differences between block cipher and stream cipher.

(b) Describe the basic use of hash function.

7. Explain DES algorithm.

8. Find the moment of inertia of a thin uniform rod of length $2a$ about the line through its one extremity perpendicular to the rod.

Part—C

(Long Answer Type Questions)

Answer *any four* questions of the following :

4×10

9. Solve the following linear programming problem by simplex method :

$$\text{Min } Z = x_1 - 3x_2 + 2x_3$$

subject to

$$3x_1 - x_2 + 2x_3 \leq 7$$

$$-2x_1 + 4x_2 \leq 12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0$$

10. Show that in the catenary $y = c \cosh \frac{x}{c}$

(a) the length of the arc being measured from the vertex to any point is given by $y = c \sinh \frac{x}{c}$;

(b) $y^2 = c^2 + s^2$, the arc being measured from the vertex.

11. Explain RSA algorithm and give example of generation of public and private key.

12. The cardioid $r = a(1 + \cos \theta)$ revolves about the initial line. Find the surface and volume of cardioid.

13. (a) Find the orthogonal trajectories of the curve

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1$$

where λ being the parameter of the family.

(b) Solve :

$$y = px + p - p^2$$

14. Solve :

(i) $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = \sin x + \cos x$

(ii) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^x$
